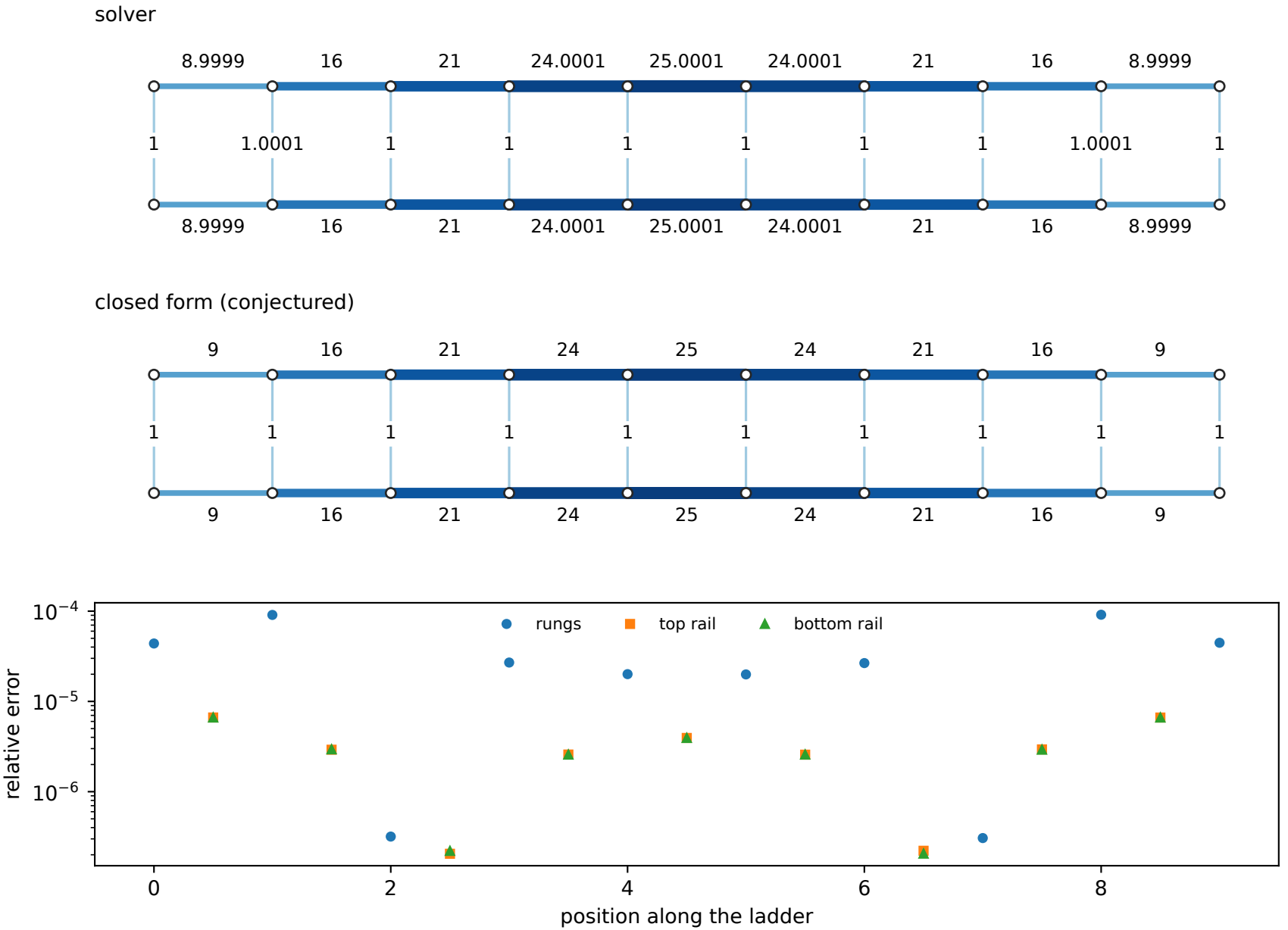n = 10: $\lambda_2^\star$ = 0.00588235 closed form 0.00588235 max error: rails 6.6e-06, rungs 9.1e-05